Corrosion control

It is impractical to eliminate corrosion but corrosion can be controlled and minimized by several methods. The rate of corrosion can be controlled by either modifying the base metal/alloy or the environment. Following methods are adopted for protection of metal.

Anodic Protection

1. **Protective Coatings**: Protective coatings are used to isolate the cathode and anode regions. The coating may be metallic, non-metallic or organic substances. The coated surface isolates the metal from the corroding medium. The coating applied must be chemically inert towards the environment.

A. Metallic Coatings: Metallic coatings are mostly applied on Iron and steel because these are cheap and commonly used construction materials. There are two types of metallic coatings.

(i)Anodic coatings:

The base metal which is to be protected is coated with a more anodic metal for eg. Coatings of Zn, Al and Cr on steel, are anodic because their electrode potentials are lower than that of the base metal ie. Fe. under atmospheric factors corrosion of more active metal layer occurs and base metal remains protected. eg. galvanized Iron sheet. (Coating of Zn on Iron)

(ii) Cathodic Coatings:

It is obtained by coating a more inert metal having higher electrode potential than the base metal. eg. Coating of Sn, Cd, Ni on Fe surface. The coating should be continuous and free from pores and cracks. These coating metals usually have higher corrosion resistance than the base metal.

If the coating of these noble metals breaks or cracks down, then localized corrosion starts due to formation of galvanic cell, hence it should be smooth & non-porous.
eg. tinning of Iron.

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Methods of Application Of Metallic Coating:

I. Hot Dipping

It is used for producing a coating of low melting metal such as Zn, Sn, Pb, Al etc on relatively higher melting metals such as iron, steel, copper etc. This is done by immersing the base metal covered by a layer of molten flux. The flux is used to keep the base metal surface clean and also to prevent oxidation of the molten metal. Most widely used hot dipping methods are : (i) galvanization and (ii) tinning

(i)Galvanization:

It is the process of coating Zn over iron or steel sheet by immersing it in molten Zn. The

procedure involves the following stages.

The iron or steel article is first cleaned by pickling with dil H_2SO_4 for 15-20 min.at 60 - 90°C in an acid bath. This treatment also removes any oxide layer present on the surface of the metal. The article is then washed with water in a washing bath & dried in a drying chamber. It is then dipped in a bath of molten Zn kept at 425 –435°C. The Surface of the bath is covered with NH_4Cl flux to prevent oxide formation. The article gets coated with a thin layer of Zn. It is then passed through a pair of hot rollers to remove excess of Zn and to get uniform thickness for coating. Then it is annealed at about 650°C & cooled slowly. In the case of Zn coating even if the protecting layer has cracks on it, iron being cathodic does not

get corroded.

Applications: This method is widely used for protection of Fe from atmospheric corrosion in the form of articles like roofing sheets, wires, pipes nails, screws, tubes etc. It is to be noted that galvanized utensils should not come in contact with acids.

(ii)Tinning:

It is the process of coating Sn over Fe or steel articles by immersing it in molten Sn. The

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process consists in treating the iron sheet with dil. H_2SO_4 to remove any oxide film. After this it is passed through a bath of $ZnCl_2$ flux which helps the molten Sn to adhere to the metal sheet.

Applications: Tinning is widely used for coating steel, Cu and brass sheets which are used for making containers for storing food studs, oils, kerosene & packing food materials. Tinned Cu sheets are used for making cooking utensils & refrigeration equipments.

II. Electro-plating or Electro-deposition: It is probably the most important and most frequently applied industrial method of producing metallic coatings. Electroplating is carried out by a process called electrolysis. Thus in this process, the coating metal is deposited on the base metal by passing direct current through an electrolyte containing the soluble salt of the coating metal. The base metal to be electroplated is made the cathode of the electrolytic cell whereas the anode is either made of the coating metal itself or an inert material of good electrical conductivity like graphite.

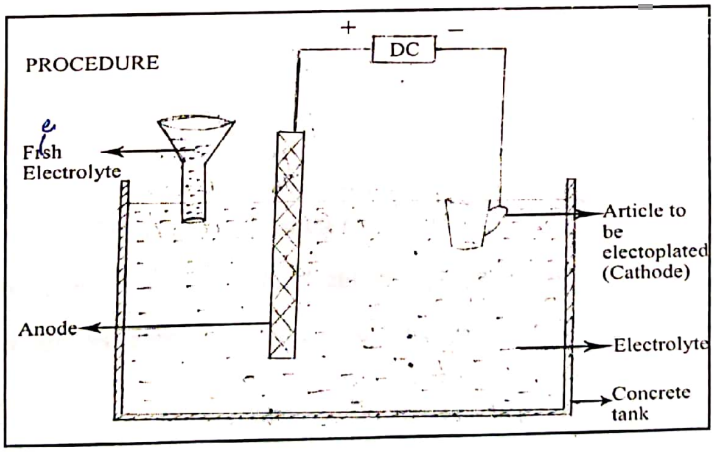


Fig 5.18: Electroplating of metal

The article has to be cleaned thoroughly before electroplating. For this, the article is first treated with organic solvents to remove oils, grease etc. This process is called solvent cleaning. Next, alkali cleaning is applied to remove old painting coatings from metal

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surfaces. Then the article is washed with dil H_2SO_4 or HCl to remove surface oxide films. The cleaned article is then made cathode of an electrolytic cell. The anode is either the coating metal itself or an inert material of good electrical conductivity like graphite. The electrolyte is a solution of a soluble salt of the coating metal. The anode and cathode are dipped in the electrolytic solution taken in a concrete tank. On passing direct current coating metal ions migrate to the cathode and get deposited there. Thus a thin layer of coating metal is obtained on the article which acts as the cathode. For electroplating of Ni, $NiSO_4$and $NiCl_2$ are used as the electrolyte. For electroplating of Cr, chromic acid is used as the electrolyte. For Au plating, $AuCl_3$ solution is taken as the electrolyte.

III.Metal spraying

In this process, the coating metal in the molten state is sprayed on the previously cleaned base metal with the help of a sprayer. The sprayer coatings are continuous but somewhat porous sealer oil is applied on such a coating to provide a smooth surface. However, adhesion strength of metallic spraying is usually lesser than that obtained by hot dipping or electroplating. It is therefore essential to have a cleaned metal surface.

Applications:

1.Coating can be applied to fabricated structure

2.Coating can be applied to large & irregular shaped articles.

3. Coating can be applied even on to non – metallic bases made of glass, wood, plastic.

IV.Metal Cladding

In this process, a thick homogeneous layer of coating metal is bonded firmly & permanently to the base metal on one or both the sides. This method Increases corrosion resistance. The choice of cladding material depends on the corrosion resistance required for any particular environment. Nearly all existing corrosion resisting metals like Ni, Cu, Al, Ag, Pt and alloys

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like stainless steel, Ni alloys, Cu alloys can be used as cladding materials. Cladding can be done by different means.

a. Fusing cladding material over the base metal

b. Welding

c. Rolling sheets of cladding material over base metal.

Generally cladding is done by arranging thin sheets of the coating metal and the base metal sheet in the form of a sandwich which are then passed through rollers under the action of heat & pressure. For aircraft applications, sheets of duralumin are sandwiched between 2 layers of pure Al.

[Alloy of Al, Cu, Mn, Mg SiO_2]

V. Cementation: In this method, metal to be coated forms an alloy with the surface of base metal. The article to be coated is packed in powdered coating metal in a rotating drum at a temperature slightly below the melting point of more fusible metal.

Application: The process is used for coating a thin uniform film on screens, nuts, bolts etc.

B. Non-Metallic Coatings: These may include coating of porcelain, clay etc. or phosphate coating or coating of organic paints, varnishes, lacquers etc. These coatings provide both prevention and smoothness to metallic surface.

2. Cathodic Protection: The principle involved in this method is to force the metal to behave like cathode thereby corrosion does not occur. It may be carried out in two ways:

a. Sacrificial anodic protection method:

- The metallic structure to be protected is connected by a wire to a more anodic metal like Zn, so that all corrosion is concentrated at this more active metal.
- The more active metal itself corroded slowly while the metallic structure is protected.

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- The more active metal is called "sacrificial anode" and it is replaced by fresh one, when consumed completely.
- Buried pipelines, underground cables, marine structures, ship-hulls, water-tanks, pipes etc are protected by this method.

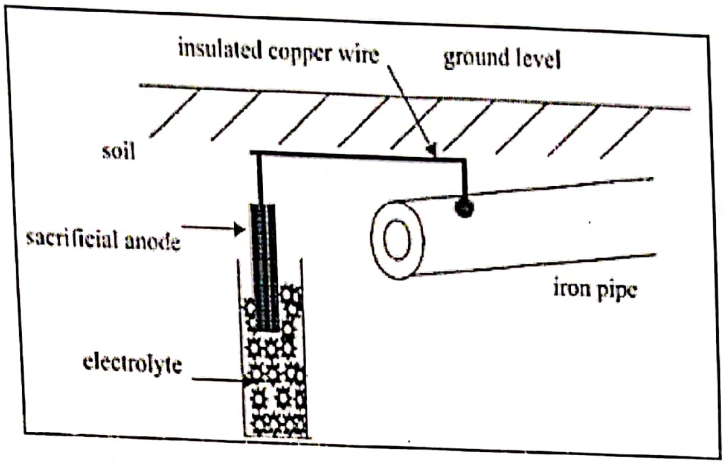


Fig 5.19: Sacrificial ~~Cathodic~~ anodic Protection

b. Impressed Current Cathodic Protection:

- An impressed current is applied in opposite direction to nullify the corrosion current and convert the corroding metal from anode to cathode.
- Impressed current is derived from DC source (battery) with an insoluble anode(graphite).
- Anode is in backfill (composed of coke breeze or gypsum) so as to increase the electrical contact with the surrounding soil.
- A sufficient d.c current is applied to insoluble anode, buried in the soil or immersed in the corroding medium and connected to the metallic structures to be protected.
- It is useful for large structures for long-term operations.eg. buried water or oil pipes, water tanks.

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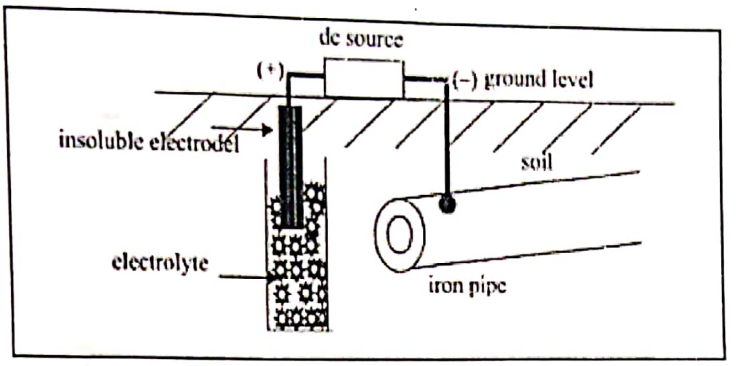


Fig 5.20: Impressed current Cathodic Protection

3. Proper Designing: The design of material plays important role in corrosion. Design of material should be such that corrosion, even if occurs, is uniform and doesn't result in localized corrosion.Some important design principles are:

i) Choice of the right type of material is the main factor for corrosion control. The selection of metal should be based on its chemical properties as well as its cost and structure.

ii) The metal used should be as pure as possible. Presence of even minute amounts of some impurities may lead to severe corrosion. For example, minute quantities of iron in magnesium or lead in zinc in die-casting alloys are detrimental.

iii) Strength and corrosion resistance may be improved by alloying. For example, stainless steel containing chromium forms an oxide film protecting it from further attack. Nimonic alloys (Ni-Cr-Mo) used in gas turbines are resistant to hot gases.

iv)Heat treatment like annealing reduces internal stress and corrosion.

v) When two metals have to be in contact, they should be selected so that their oxidation potentials are as near as possible. Further, the area of more active metal, i.e., the anode should be larger than that of the cathode or less active metal. A protective coating over the two metals

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v) When two metals have to be in contact, they should be selected so that their oxidation potentials are as near as possible. Further, the area of more active metal, i.e., the anode should be larger than that of the cathode or less active metal. A protective coating over the two metals reduces pitting.

vi) If an active metal is used, it should be insulated from the more cathodic area.

vii). 'Avoid crevices between assembled or joined materials. Welding is a better joining technique than soldering, for mechanical fastness. Galvanic cells develop in case of soldering or brazing.

viii The parts should be designed in such a way that a stagnant pool of liquid is not collected.

ix). Moisture should be excluded wherever practicable; no corrosion occurs without the presence of moisture. The assembly should be designed such that the moisture should be as low as possible. Proper aeration of metallic parts is required.

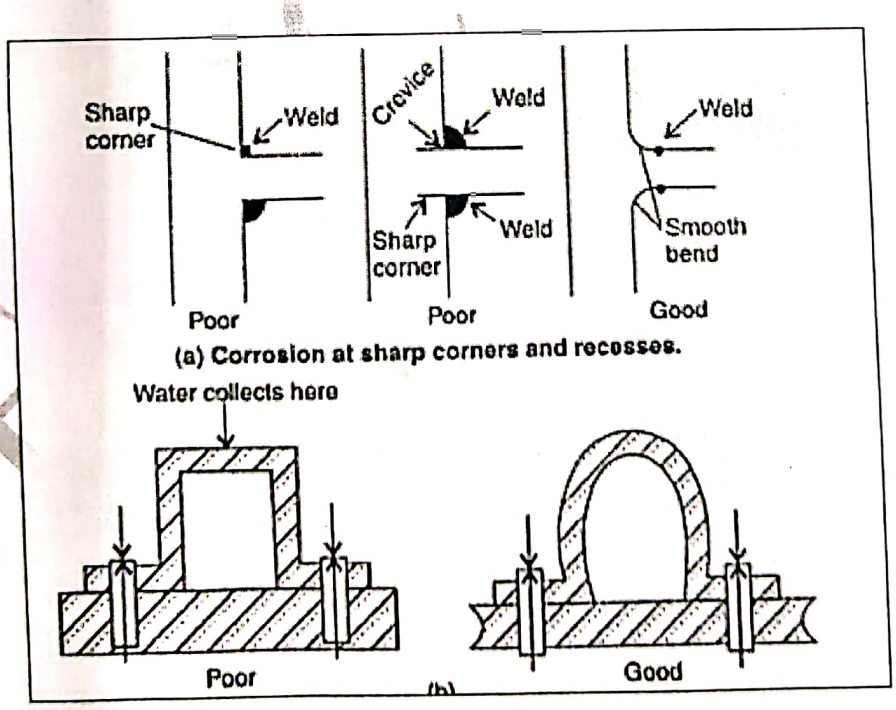


Fig 5.21a: Design of material

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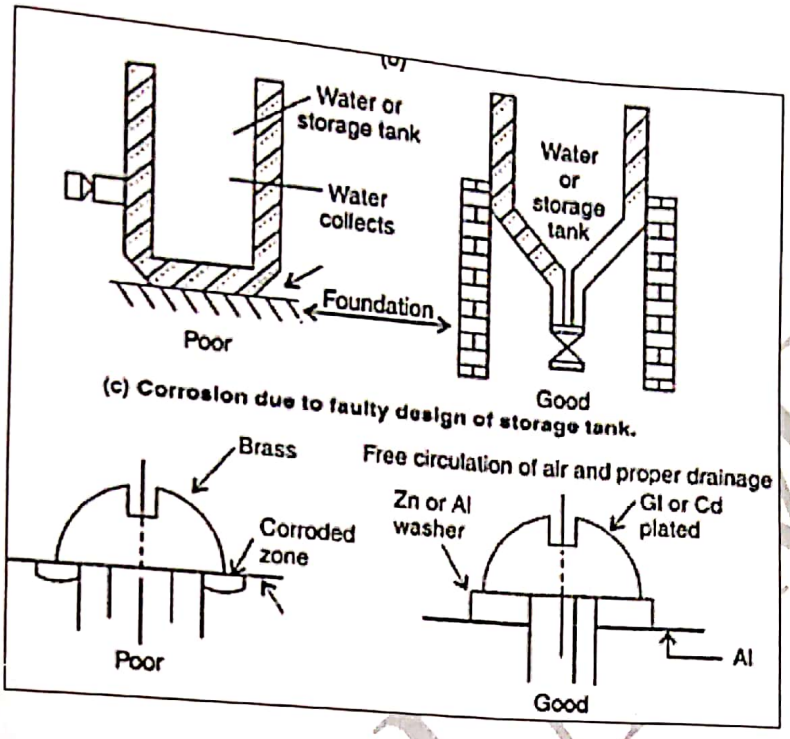


Fig 5.21b: Proper Design of material

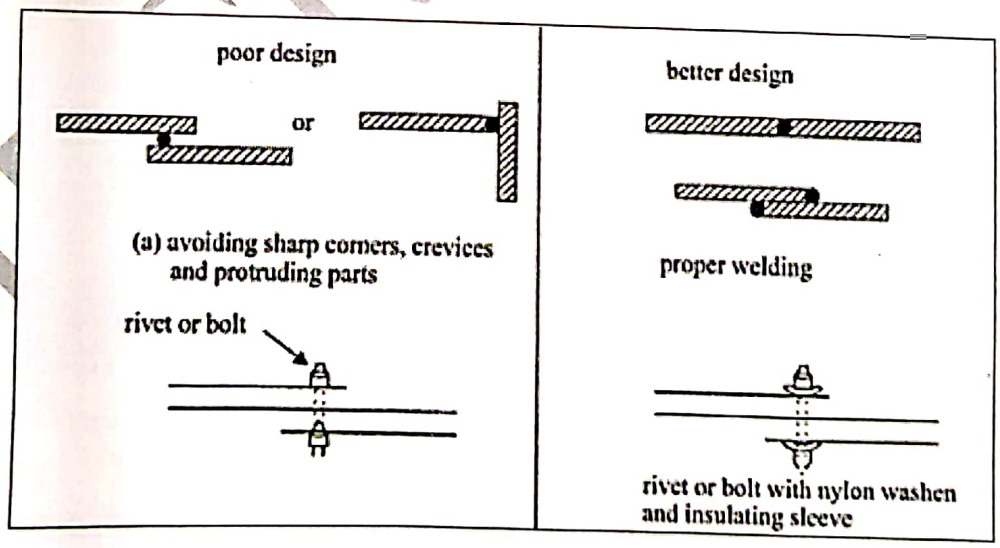


Fig 5.22: Properly welded metal plates and insulated screws

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